

# XRPL Validator — Daily Engineering Report • 2026-09-04

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Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

## Summary

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- 7 commits (xrpl-validator 7)

## Engineering changes

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- **fix(ledger): a two-asset AMM deposit mints its sized tokens, LPTokenOut is only a floor (finding 155)** ( 0f5126f , xrpl-validator) On the tfTwoAsset path rippled sizes the deposit from Amount/Amount2 and the pool's balances — equalDepositLimit picks the leg whose proportional counterpart fits, rounds the tokens ( getRoundedLPTokens ) and mints exactly those — and treats an LPTokenOut on the transaction only as lpTokensDepositMin: deposit() fails with tecAMM\_FAILED when the sized tokens fall below it, before anything moves. We minted the ...
- **fix(ledger): an AMM withdrawal charges reserve for a trust line it must create (finding 156)** ( 31093f7 , xrpl-validator) AMMWithdraw::withdraw runs sufficientReserve for every asset it delivers (fixAMMv1\_2): when the account holds no trust line for an IOU being withdrawn, the line it is about to create is a new owned object, and TrustSet's rule applies — nothing owed while the account owns fewer than two objects, otherwise base plus increment times OwnerCount + 1, measured against the balance before the fee ( max(priorBalance, ...
- **fix(ledger): a direct hop into a globally frozen account is dry (finding 157)** ( 21a8437 , xrpl-validator) checkFreeze (StepChecks.h:19-46) runs on every DirectStep that is not the strand's only step: the hop is terNO\_LINE when the DESTINATION account carries lsfGlobalFreeze — the holder's own flag, whatever the issuer — when the (src, dst) trust line carries the destination's freeze bit, or when either side deep-froze it. Our freeze model looked only at the issuer's global freeze and the issuer-side line flag, so a ...
- **fix(ledger): a self-cross removes every self-offer inside the limit, not just the tip (finding 158)** ( bf5936e , xrpl-validator) Ledger #106755558, tx 15B9744CE01E: rw7nJtEN places an offer selling 256225 BBRL for 50000 RLUSD (limit 5.1243 BBRL per RLUSD). The direct tip is the account's own 20000 RLUSD bid at 5.1232 and the next level its own bid at 5.1238, both inside the limit. rippled's execOffer (BookStep.cpp:439-451) removes the first through limitSelfCrossQuality and, since no offer has been attempted yet, resets the level quality ...
- **fix(ledger): a crossing pass is judged on its pool slice plus its fills (finding 159)** ( 10b3cf0 , xrpl-validator) Ledger #106755996, tx D84E769E7583: rhhh49pF's tfImmediateOrCancel offer sells 21.88 XRP for 210.047995624 CNY. The direct book's tip (rwvLbHQ, 210.048 CNY for 21880000 drops) prices exactly at the taker's limit, and the XRP/CNY pool is anchored on it: tryAMM takes 6980735 drops for 67.015056 CNY, a hair inside the limit, then execOffer fills the tip for the remaining 143.032939624 CNY at the drop ceiling, 14899265 ...

- **fix(ledger): a tfSell residual divides the gross by the rate with rippled's lossy divRound (finding 160)**  
( 102021b , xrpl-validator) Ledger #106755998, tx A3D3541FD33F: rURtT5MM's tfSell offer sells 0.12393 BTC (issuer rate 1.0015) for RLUSD and crosses one offer, debiting 0.05880101336798101 BTC gross. OfferCreate.cpp derives the rested TakerGets as takerGets minus divideRound(actualAmountIn, gatewayXferRate, asset, roundUp), and divideRound is STAmount's divRound: muldivRound ceils the 17-18 digit quotient, then canonicalizeRound truncates an ...
- **fix(node): RPC client reports the real count and time spent on a fallback endpoint ( 4e7e5b4 , xrpl-validator)**  
The periodic 'Trying primary endpoint again after N successes on fallback' line printed a counter that was never reset, so N was the process's lifetime success count (1,816,400 this afternoon) and read as hours parked on s1.ripple.com when the client had in fact been back on the the reference node primary within a second of every failover.

## XRPL research

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### Automated research — candidate amendment ideas

*Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: [iceland.halcyon-names.io](https://iceland.halcyon-names.io)). They are exploration leads, not proposals.*

- **AMM Price-Triggered Stop Offers (dex)** Add a tfTriggerOnAMM flag plus sfTriggerAMMID (AMM keylet), sfTriggerDirection, and sfTriggerPrice (STAmount) SFields to OfferCreate. A triggered offer
- **On-Ledger Travel Rule Envelope for Payments (compliance)** Add an optional sfTravelRuleEnvelope VL SField (opaque ciphertext, capped at 4KB) to Payment, plus a new AccountRoot flag lsfRequireTravelRule and pai
- **Hash-Blind Offer Tiebreak for MEV Resistance (dex)** Change the tie-break rule for equal-quality Offers landing in the same ledger's SortedDirectories-ordered book: instead of ordering ties by canonical

## Appendix — commits

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| Time  | Repo           | Commit  | Subject                                                                                                  |
|-------|----------------|---------|----------------------------------------------------------------------------------------------------------|
| 01:00 | xrpl-validator | 0f5126f | fix(ledger): a two-asset AMM deposit mints its sized tokens, LPTokenOut is only a floor (finding 155)    |
| 05:17 | xrpl-validator | 31093f7 | fix(ledger): an AMM withdrawal charges reserve for a trust line it must create (finding 156)             |
| 06:29 | xrpl-validator | 21a8437 | fix(ledger): a direct hop into a globally frozen account is dry (finding 157)                            |
| 08:30 | xrpl-validator | bf5936e | fix(ledger): a self-cross removes every self-offer inside the limit, not just the tip (finding 158)      |
| 08:57 | xrpl-validator | 10b3cf0 | fix(ledger): a crossing pass is judged on its pool slice plus its fills (finding 159)                    |
| 08:57 | xrpl-validator | 102021b | fix(ledger): a tfSell residual divides the gross by the rate with rippled's lossy divRound (finding 160) |
| 17:15 | xrpl-validator | 4e7e5b4 | fix(node): RPC client reports the real count and time spent on a fallback endpoint                       |